

BSDS V: Regression Techniques

Problems Set 3: Departure from Assumptions of Normal Linear Model

Throughout this problem set, we will consider the multiple linear regression model with response variable y and a list of regressors/covariates $\{x_1, \dots, x_p\}$. Based on n observations the model is

$$\underline{Y} = X\underline{\beta} + \underline{\varepsilon}, \quad \text{where} \quad \underline{Y} = \begin{bmatrix} Y_1 \\ \vdots \\ Y_n \end{bmatrix}, \quad X = \begin{bmatrix} 1 & \underline{x}_{(1)} & \dots & \underline{x}_{(p)} \end{bmatrix}, \quad \underline{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{bmatrix}, \quad \text{and} \quad \underline{\varepsilon} = \begin{bmatrix} \varepsilon_1 \\ \vdots \\ \varepsilon_n \end{bmatrix}. \quad (1)$$

We assume that $E(\underline{\varepsilon}) = \underline{0}$ and $\text{var}(\underline{\varepsilon}) = \sigma^2 I_n$.

1. Let the underlying actual model be (1). However, you have collected data on q additional variables $\{z_1, \dots, z_q\}$ and work on the model:

$$\underline{Y} = X\underline{\beta} + Z\underline{\gamma} + \underline{\varepsilon}^*, \quad \text{where} \quad \underline{\varepsilon}^* \sim N(\underline{0}, \sigma^2 I), \quad Z = \begin{bmatrix} \underline{z}_{(1)} & \dots & \underline{z}_{(q)} \end{bmatrix} \quad (2)$$

Consider a new observation $(\underline{x}_0^\top : \underline{z}_0^\top)$ with corresponding response Y_0 . Compare the confidence interval of $\underline{x}_0^\top \underline{\beta}$ obtained from your model (2) with the one obtained from the actual model (1). Also, compare the prediction interval for Y_0 obtained from model (2) with the one obtained from the actual model (1).

2. Consider the linear model in (1) as your working model. Suppose, however, that the underlying true model is as in (1) except for the fact that $\text{var}(\underline{\varepsilon}) = \sigma^2 \text{Diag}(t, t^2, \dots, t^n)$ for some number $t > 1$. How would this misspecification affect the variance of OLS $\hat{\underline{\beta}}_{\text{OLS}}$ and expectation of $\hat{\sigma}_{\text{UE}}^2 = \text{RSS}/(n-p-1)$?
3. Consider the above setup in Q.2. However, suppose that you know t . How would you modify the regression estimate of $\underline{\beta}$ so that its variance is smaller than the variance of $\hat{\underline{\beta}}_{\text{OLS}}$ obtained in Q.2 (in the sense that the difference of the two covariance matrices is negative semi-definite).
4. Let $\underline{x}_{(1)}, \dots, \underline{x}_{(n)}$ be n observations on p -covariates, i.e., $\underline{x}_{(i)} \in \mathbb{R}^p$, and define $\bar{\underline{x}} = n^{-1} \sum_{i=1}^n \underline{x}_{(i)}$, and $S_X = n^{-1}(\underline{x}_{(i)} - \bar{\underline{x}})(\underline{x}_{(i)} - \bar{\underline{x}})^\top$. Consider the Mahalanobis distances:

$$\frac{\text{MD}_i}{\text{MD}_{i,j}} = \frac{\left\{ (\underline{x}_{(i)} - \bar{\underline{x}})^\top S_X^{-1} (\underline{x}_{(i)} - \bar{\underline{x}}) \right\}^{1/2}}{\left\{ (\underline{x}_{(i)} - \underline{x}_{(j)})^\top S_X^{-1} (\underline{x}_{(i)} - \underline{x}_{(j)}) \right\}^{1/2}} \quad \left| \begin{array}{l} \text{MD of } \underline{x}_{(i)} \text{ from the center,} \\ \text{MD between } \underline{x}_{(i)} \text{ and } \underline{x}_{(j)}. \end{array} \right.$$

Suppose $\underline{x}_{(i)}$ s are i.i.d. samples from a p -variate distribution with finite variance-covariance matrix.

- (a) Show that $E(\text{MD}_i^2) = p$.
- (b) Show that $E(\text{MD}_{i,j}^2) = 2np/(n-1)$.

- (c) Define $Z = \begin{bmatrix} \underline{x}_{(1)}^\top \\ \vdots \\ \underline{x}_{(n)}^\top \end{bmatrix}$, $X = \begin{bmatrix} 1 & Z \end{bmatrix}$ and $P(X) = X(X'X)^{-1}X' = ((p_{i,j}))$. Using (i) and (ii), derive the expectations of $p_{i,i}$ and $p_{i,j}$ for $i \neq j$.

5. Consider fitting a full rank multiple regression model $\underline{y} = X\underline{\beta} + \underline{\epsilon}$ by the least squares method. Let $P(X) = ((p_{ij}))$ be the orthogonal projection matrix onto $\mathcal{C}(X)$ and $P_{(i)} = ((p_{jk(i)}))$ be the same when the i^{th} observation was removed. Let $\underline{x}_i$ be the i^{th} row of X (as a column vector), e_i be the i^{th} residual from the full model fit, and $\hat{\underline{\beta}}_{\sim(i)}$ be the i^{th} deleted OLS estimate of $\underline{\beta}$.

(a) Show that $e_{i(i)} = \frac{e_i}{1 - p_{ii}}$, where $e_{i(i)} = y_i - \underline{x}_i^T \hat{\underline{\beta}}_{\sim(i)}$.

(b) Prove that

$$p_{jk(i)} = p_{jk} + \frac{p_{ik}p_{ij}}{1 - p_{ii}}. \quad [\text{You may assume } \max\{j, k\} < i]$$

(c) In case of no intercept model, show that $-0.5 \leq p_{ij} \leq 0.5$, $i \neq j$.

6. Consider fitting a regression model with regression matrix X , response vector $\underline{Y}$ and unknown parameter vector $\underline{\beta}$. Let $\hat{\underline{\beta}}$ be the OLS estimate of $\underline{\beta}$.

Suppose that a new observation $(\underline{x}_{n+1}, Y_{n+1})$ is added. Show that the residual sum of squares is increased by an amount $\frac{e_{n+1}^2}{1 + \underline{x}'_{n+1}(X'X)^{-1}\underline{x}_{n+1}}$, where $e_{n+1} = Y_{n+1} - \underline{x}'_{n+1}\hat{\underline{\beta}}$.